
GIT PUSH THE BEATS

Melody Dive Use-Case Specification: Sign Up

Version 1.0

Melody Dive	Version: 1.0
Use-Case Specification: Sign Up	Date: 15/11/2025
UC SignUp	

Revision History

Date	Version	Description	Author
15/11/2025	1.0	Add: add details information about this use case	Thái Minh Huy

Melody Dive	Version: 1.0
Use-Case Specification: Sign Up	Date: 15/11/2025
UC SignUp	

Table of Contents

1. Use-Case Name: Sign Up	4
1.1 Brief Description	4
2. Flow of Events	4
2.1 Basic Flow	4
2.2 Alternative Flows	5
2.2.1 Username Already Exists	5
2.2.2 Invalid Input	5
3. Special Requirements	5
3.1 Security	5
3.2 Uniqueness	5
3.3 Validation	5
3.4 Performance	5
3.5 Efficiency	6
4. Preconditions	6
4.1 User Status	6
4.2 Account Status	6
5. Postconditions	6
5.1 Account Creation	6
5.2 Redirection	6
6. Extension Points	6
6.1 Login	6

Melody Dive	Version: 1.0
Use-Case Specification: Sign Up	Date: 15/11/2025
UC SignUp	

Use-Case Specification: Sign Up

1. Use-Case Name: Sign Up

1.1 Brief Description

The “Sign Up” use case allows an unregistered user to create a new account on the system by providing required information such as “Username”, “Full Name”, “Email Address”, “Password”, “Confirm Password”. Once successfully registered, the user can proceed to log in.

2. Flow of Events

2.1 Basic Flow

1. The user selects the “Sign Up” option from the login page.
2. The system displays a registration form requiring:
 - Username.
 - Fullname.
 - Email.
 - Password.
 - Confirm Password.

Figure 1: The interface of use case Sign Up

3. The user fills in all required fields and submits the form.
4. The system validates the input.
 - Ensure the username is not already taken.
 - Checks password requirements such as minimum length.
5. If all data is valid:
 - The system creates a new user account.
 - Displays a success message and redirects the user to the login page.

Melody Dive	Version: 1.0
Use-Case Specification: Sign Up	Date: 15/11/2025
UC SignUp	

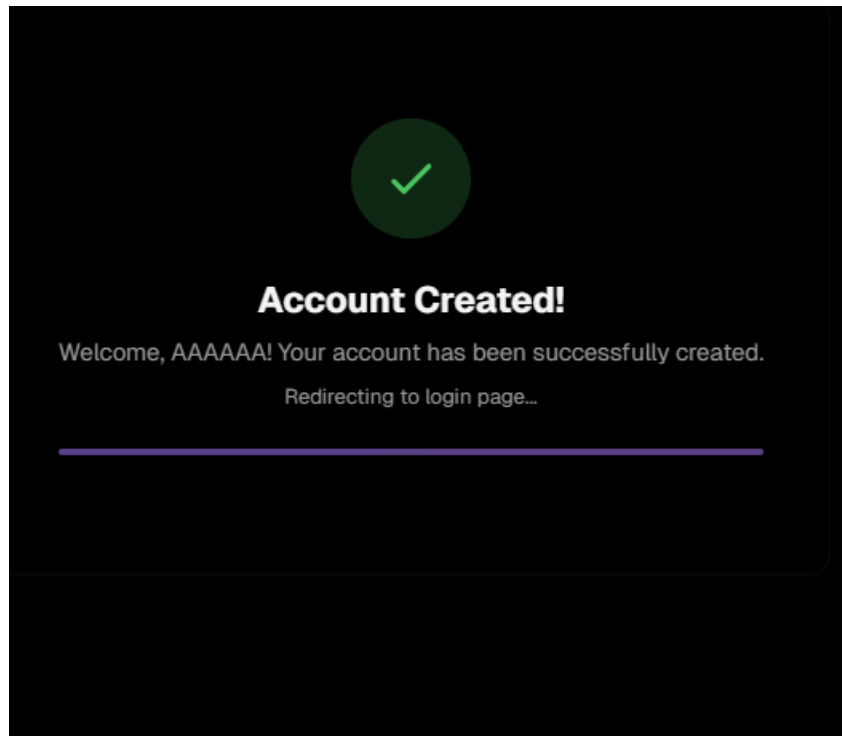


Figure 2: The interface after signing up completed

6. The user can now log in with the registered credentials.

2.2 Alternative Flows

2.2.1 Username Already Exists

If the entered username already exists:

- The system displays an error message: “Username already taken”.
- The user is required to enter a new username

2.2.2 Invalid Input

If required fields are missing or invalid:

- The system highlights the issue such as “Password too short”.
- The user must correct the input and resubmit.

3. Special Requirements

3.1 Security

The system must be able to check the password’s quality. A password must have at least 6 characters, have a number, one special character in the password.

3.2 Validation

Email format should be validated (if provided) with a success rate of 98% accuracy.

3.3 Performance

Form validation should occur both client-side and server-side, completing within 3 seconds.

Melody Dive	Version: 1.0
Use-Case Specification: Sign Up	Date: 15/11/2025
UC SignUp	

3.4 Efficiency

Registration must complete within 5 seconds under normal load conditions.

4. Preconditions

4.1 User Status

The user is not currently logged in.

5. Postconditions

5.1 Account Creation

A new account is successfully created in the database

5.2 Redirection

The user is redirected to the login page for authentication

6. Extension Points

6.1 Login

After successful sign-up, the user can proceed to log in using the newly created credentials.